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|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 6(a) | <p>I maintains length of chromosomes / prevents chromosome shortening</p> <p><i>any two from:</i></p> <p>permit continued replication ;</p> <p>prevent loss of genes ;</p> <ul style="list-style-type: none"> <li>I prevents gene damage genetic material is neutral</li> <li>A genetic / coded, information <i>for genes</i></li> <li>A information on DNA</li> <li>A protein coding regions of DNA</li> <li>A exons</li> </ul> <p>protect ends of chromosomes from being, degraded / AW ;</p> <p>AVP ;</p> <p>e.g. prevents ends of chromosomes from being attached to each other<br/>not mistaken for a break in DNA that needs repairing</p>                                                                                                                                                                                                                                                                                                                | 2 |
| 6(b) | <p><i>Idea that</i>, mitosis / cell cycle / (DNA) replication / (cell) division, occurs in both cell types (continuously / regularly / AW) ;</p> <ul style="list-style-type: none"> <li>R uncontrolled mitosis</li> <li>R meiosis</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1 |
| 6(c) | <p>induced fit ;</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1 |
| 6(d) | <p><i>allow low(er) rate for <math>0.2 \mu\text{mol dm}^{-3}</math> and high(er) rate for <math>1.6 \mu\text{mol dm}^{-3}</math></i></p> <p><i>any two from:</i></p> <p>comparison in terms of <u>active site(s)</u> ;</p> <p>e.g. (low rate) active sites, not all occupied / spare / not all saturated</p> <p><b>or</b></p> <p>(high rate) active sites all occupied / no spare active sites / active sites saturated bod 'active sites are limited'</p> <p>comparison in terms of enzyme-substrate complexes formed (per unit time)</p> <p>e.g. (low rate) few(er) ES complexes formed</p> <p><b>or</b></p> <p>(high rate) many / more, ES complexes formed ;</p> <ul style="list-style-type: none"> <li>A (low rate) few(er) collisions between enzyme and substrate ora</li> <li>(low rate) substrate concentration limiting factor</li> </ul> <p><b>or</b></p> <p>(high rate) limiting factor is, enzyme concentration / not substrate concentration;</p> | 2 |